

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _____ 129

Does R2 show up as an OSPF neighbor on R1? _____ yes

Does R2 show up as an OSPF neighbor on R3? _____ no

What does this information tell you?

All traffic from R3 destined for the 192.168.2.0/24 network will be directed through R1.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_____ R2(config)# router ospf 1

_____ R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

_____ 65 = 64 + 1

_____ 64 for T1 serial link (1.544Mb/s) and 1: R2 Gigabit 0/0 LAN link

Is R2 listed as an OSPF neighbor to R3? _____ Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
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Why would you want to change the OSPF default reference-bandwidth?

_____ a greater default reference bandwidth setting is required to achieve a more precise cost calculation for the faster links _____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

_____ 192.168.3.0/24: $781+1=782$ (R1 S0/0/1 + R3 G0/0)

_____ 192.168.23.0/30: $781+64=845$ (R1 S0/0/1 and R3 S0/0/1)

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

_____ 1562 _____

_____ the route to the 192.168.23.0/24 network utilizes two serial links, each 781 _____

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF will select the route with the lowest total cost. The most cost-effective route is via R1-S0/0/0/R2-S0/0/1, and R3-G0/0, which totals $781 + 781 + 1$, resulting in an accumulated cost of 1,563. In comparison, the cost of the route through R1-S0/0/1 and R3-G0/0 is $1,565 + 1$, yielding an accumulated cost of 1,566.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

___In OSPF, the router ID is designated to manage the Designated Router (DR) and Backup Designated Router (BDR). Properly controlling the assignment of the router ID is crucial for maintaining predictable and stable routing behavior, helping to avoid unexpected changes caused by interface reassessments or router restarts._____

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2. Why is the DR/BDR election process not a concern in this lab?

___The DR/BDR election process is relevant only in multi-access networks like Ethernet or Frame Relay. Since the serial links used in this lab are point-to-point, there is no need for a DR/BDR election.

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3. Why would you want to set an OSPF interface to passive?

___It can conserve bandwidth by removing unnecessary OSPF information and preventing OSPF from sending hello packets and establishing adjacencies on an interface, all while still advertising the network.

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

___ Utilizing areas in OSPF facilitates a hierarchical network architecture, which helps lower routing overhead and enhances scalability. _____

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5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

___ network 10.0.0.0 0.255.255.255 area 0 _____

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6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

___ The ~~OSPF~~ route will be selected because it has a lower administrative distance of 110, compared to RIP's 120 and ~~EIGRP~~'s internal distance of 90 and external distance of 170, making it the preferred route.

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